

Course : Data Science

Exercise No: Advanced SQL NULL Functions

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Question 1 : Display salary and replace NULL with 0

Syntax

SELECT employee-id,
name,

IF NULL (Salary, 0) AS Salary-with-default

FROM Employees;

Table

employeeid	name	Salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Question 2 - Display department and replace NULL with 'Not Assigned'

Syntax

SELECT employee-id,
name,

IF NULL (department, 'Not Assigned')

AS department-name

FROM Employees;

(2)

table	name	department - name
employee-id	Alice	HR
1	Bob	IT
2	Charlie	NOT Assigned
3	Dana	Finance
4		

Q3 Find orders with NULL Customer_id using ISNULL

Syntax

SELECT order-id,
Customer_id

FROM Orders

WHERE ISNULL (Customer_id);

Table

Order-id	Customer-id
103	NULL

Q4 Replace Null delivery dates with Pending

Syntax

SELECT order-id,
Customer-id,

IF NULL (delivery-date, 'Pending') AS delivery
Status

FROM Orders;

3

table

Order-id	customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	Null	2024-12-03

Q5. Replace NULL grades with 0

Syntax
 SELECT Student-id,
 name,
 IFNULL(grade, 0) AS Final-grade
 FROM Students;

Table

Student-id	name	Final-grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6 Count Students without grades.

Syntax
 SELECT Count(*) AS not-graded-Count
 FROM Students
 WHERE ^{grade} IS NULL;

table

not-graded-Count
1

Q7. Calculate final price after discount

Syntax
 SELECT product-id, name, Price - IFNULL
 (discount, 0) AS Final-price
 FROM products;

table

product-id	name	final-price
501	keyboard	25
502	Mouse	10
503	Monitor	100

Q8 Count customers with no email

Syntax
SELECT COUNT (*) AS missing_email_count
FROM Customers
WHERE email IS NULL;

table

missing_email_count
2

Q9 Display email or 'No email'

Syntax
SELECT customer_id,
name,
IFNULL(email, 'No email') AS email-
display
FROM Customers;

table

Customer_id	name	email-display
1	Linda	No email
2	Joseph	joseph@mail.com
3	Nia	No email

Q10 Replace Null payment method with 'unknown'

Syntax
 SELECT payment-id,
 IFNULL(method, 'unknown') AS method-display
 Status
 FROM Payments;
 table

payment-id	method-display	Status
301	Credit	NULL
302	Paypal	Success
303	unknown	Failed

Q11 Replace Null Quantities with 0

Syntax
 SELECT item-id,
 (item-name),
 IFNULL(quantity, 0) AS quantity-checked
 FROM Inventory;
 table

item-id	item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12 Show first available reward

Syntax
 SELECT emp-id,
 COALESCE(bonus, commission, 0) AS first-
 available-reward
 FROM Employees_extra;

(2)

table	First - available - reward
emp_id	
1	300
2	100
3	0

Q13 Count Classes without rooms

Syntax:
 SELECT COUNT(*) AS no_room_Count
 FROM Classes
 WHERE room ISNULL;

table.

no_room_Count
2

Q14 Replace Null attendance Status 'Not marked'

Syntax
 SELECT Student_id
 date,
 IFNULL (Status, 'Not marked') AS attendance
 Status
 FROM Attendance;

Table.

Student_id	date	attendance - status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

Q15 Display account type and balance default

Syntax
 SELECT account_id,
 IFNULL(account_type, 'unknown') As type,
 display,
 IFNULL(balance, 0) As balance-checked
 FROM Bank-accounts;

table

account_id	type-display	balance checked
A1	Savings	0
A2	Current	5000
A3	unknown	2000

Q16 Replace Null start dates with TBD

Syntax
 SELECT project_id,
 title,
 IFNULL(start-date, 'TBD') As ~~start~~ start display
 FROM Projects;

table

Project-id	title	start-display
1	website revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Q17 Replace Null comment and rating

Syntax

```
SELECT review-id,
       Product-id,
       IF Null (Comment, 'No comment') As comment-
       display,
       IF Null (rating, 0) As rating-display
FROM Reviews;
```

table

review-id	Product-id	comment-display	rating-display
1	501	Great Product	4
2	502	No comment	0
3	503	Works fine	3

Q18 Display Supplier contact number

Syntax

```
SELECT Supplier-id,
       name,
       COALESCE (Phone, alt-phone, 'No Contact')
       AS Contact-number
FROM Suppliers;
```

table

Supplier id	name	Contact-number
1	Global Goods	123 456 789
2	Best Suppliers	987 654 321
3	ValueSource	No Contact

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Q19 Replace all user settings Null values

Syntax

```
SELECT user_id,  
IFNULL(theme, 'Light') AS theme-set,  
IFNULL(language, 'English') AS language-set,  
IFNULL(timezone, 'UTC') AS time-zone-set  
FROM user-settings;
```

table

user_id	theme-set	language-set	time-zone
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Q20 Display maintenance log with defaults.

Syntax

```
SELECT record-id,  
machine-id,  
IFNULL(issue, 'Unknown issue') AS issue-log,  
IFNULL(technician, 'Not assigned') AS technician  
name
```

FROM Maintenance;
table

record-id	machine-id	issue-log	technician name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex